

All for one and none for all

Compiling polymorphic relations without monomorphization

Dmitri Volkov, Yafei Yang, Chung-chieh Shan

2026 August 24

Introducing semiringKanren

- Goals

- Computing relations

- Types

Polymorphic semiringKanren

- Sum swap

- Expanding relations

- Edge cases and compilation

miniKanren vs. semiringKanren

miniKanren

- ▶ evaluation via stream search (top-down)
- ▶ dynamically typed (s-expressions)
- ▶ relations are unweighted (usually)

semiringKanren

- ▶ evaluation via multidimensional array operations (bottom-up)
- ▶ statically typed (finite algebraic data types)
- ▶ relations are weighted

Bottom-up polymorphism?

Polymorphism: using the same code to operate on multiple types, here using *type variables*.

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semiringKanren represents relations as finite multidimensional arrays.

Polymorphic relations need to represent arbitrarily many specifically-typed relations.

Bottom-up polymorphism?

Polymorphism: using the same code to operate on multiple types, here using *type variables*.

semiringKanren represents relations as finite multidimensional arrays.

Polymorphic relations need to represent arbitrarily many specifically-typed relations.

How to do both?

All for one and none for all

Represent all concrete
instances of a
polymorphic relation

using only a single
concrete instance

to compile away type
variables

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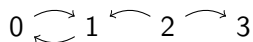
Polymorphic semiringKanren

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Representing relations



```
(defrel (graph
  (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

		to			
		0	1	2	3
from	0	0	1	0	0
	1	1	0	0	0
	2	0	1	0	1
	3	0	0	0	0

Represent graphs as relations or as adjacency matrices.
semiringKanren computes matrices for arbitrary relations!

Primitive relations

$(== \ x \ y)$

		y			
		0	1	2	3
x	0	1	0	0	0
	1	0	1	0	0
	2	0	0	1	0
	3	0	0	0	1

$(\neq \ x \ y)$

		y			
		0	1	2	3
x	0	0	1	1	1
	1	1	0	1	1
	2	1	1	0	1
	3	1	1	1	0

disj is addition

```
(defrel (r0 (x : (Num 4)) (y : (Num 4)))
  (disj
    (== x 0)
    (== y 0)))
```

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

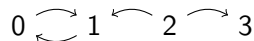
conj is multiplication

```
(defrel (r1 (x : (Num 4)) (y : (Num 4)))
  (conj
    (== x 0)
    (== y 0)))
```

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

fresh is summation

Vertices with incoming edges.



```

(defrel (incoming-edge-vertices (y : (Num 4)))
  (fresh (x : (Num 4))
    (graph x y)))
  
```

$$\sum_x \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 1 \end{bmatrix}$$

Computing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Computing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Computing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Computing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Computing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3)))))
```

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Computing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



Types

semiringKanren values inhabit finite algebraic data types.

		left	right
(left 0) :	(Sum (Num 2) (Num 3))	$\begin{bmatrix} 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$
(left 1) :	(Sum (Num 2) (Num 3))	$\begin{bmatrix} 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$
(right 0) :	(Sum (Num 2) (Num 3))	$\begin{bmatrix} 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$
(right 1) :	(Sum (Num 2) (Num 3))	$\begin{bmatrix} 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$
(right 2) :	(Sum (Num 2) (Num 3))	$\begin{bmatrix} 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$

(semiringKanren also supports unit and product types)

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Sum swap

```
(defrel (sum-swap-3-3
  (x : (Sum (Num 3) (Num 3)))
  (y : (Sum (Num 3) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 3))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Sum swap

```
(defrel (sum-swap-3-3
  (x : (Sum (Num 3) (Num 3)))
  (y : (Sum (Num 3) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 3))
      (conj
        (== x (right b))
        (== y (left b))))))
```

0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1
1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0

 $x \mapsto (\text{left } 0)$ $y \mapsto (\text{right } 0)$

Sum swap

```
(defrel (sum-swap-3-3
  (x : (Sum (Num 3) (Num 3)))
  (y : (Sum (Num 3) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 3))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$x \mapsto (\text{right } 2)$

$y \mapsto (\text{left } 2)$

Sum swap

```
(defrel (sum-swap-3-3
  (x : (Sum (Num 3) (Num 3)))
  (y : (Sum (Num 3) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 3))
      (conj
        (== x (right b))
        (== y (left b))))))
```

0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1
1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0

Sum swap

```
(defrel (sum-swap-3-3
  (x : (Sum (Num 3) (Num 3)))
  (y : (Sum (Num 3) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 3))
      (conj
        (== x (right b))
        (== y (left b))))))
```

0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1
1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0

Larger sum swap

```
(defrel (sum-swap-3-4
  (x : (Sum (Num 3) (Num 4)))
  (y : (Sum (Num 4) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 4))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Larger sum swap

```
(defrel (sum-swap-3-4
  (x : (Sum (Num 3) (Num 4)))
  (y : (Sum (Num 4) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 4))
      (conj
        (== x (right b))
        (== y (left b))))))
```

0	0	0	0	1	0	0
0	0	0	0	0	1	0
0	0	0	0	0	0	1
1	0	0	0	0	0	0
0	1	0	0	0	0	0
0	0	1	0	0	0	0
0	0	0	1	0	0	0

$x \mapsto (\text{left } 0)$

$y \mapsto (\text{right } 0)$

Larger sum swap

```
(defrel (sum-swap-3-4
  (x : (Sum (Num 3) (Num 4)))
  (y : (Sum (Num 4) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 4))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$x \mapsto (\text{right } 3)$

$y \mapsto (\text{left } 3)$

Larger sum swap

```
(defrel (sum-swap-3-4
  (x : (Sum (Num 3) (Num 4)))
  (y : (Sum (Num 4) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 4))
      (conj
        (== x (right b))
        (== y (left b))))))
```

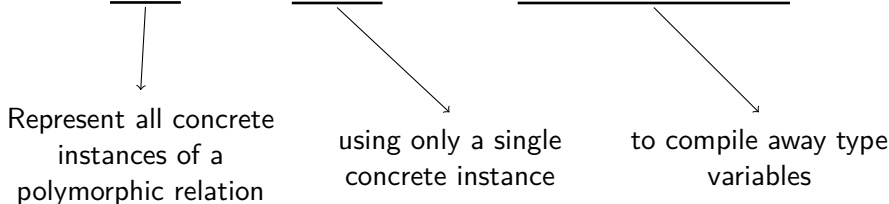
$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Polymorphic sum swap

```
(defrel (sum-swap
  (x : (Sum  $\alpha$   $\beta$ ))
  (y : (Sum  $\beta$   $\alpha$ )))
  (disj
    (fresh (a :  $\alpha$ )
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b :  $\beta$ )
      (conj
        (== x (right b))
        (== y (left b))))))
```

[?]

All for one and none for all



Expansion

semiringKanren computes a small array, and *expands* it.

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Each expanded array entry adopts weight from a smaller array entry with the same *equality pattern*.

semiringKanren uses this to *compile away* type variables.

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Fails for:

$$x \mapsto (\text{left } _ \alpha)$$

$$y \mapsto (\text{left } _ \beta)$$

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Fails for:

$$x \mapsto (\text{right } _ \beta)$$

$$y \mapsto (\text{right } _ \alpha)$$

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Succeeds for:

$$\left. \begin{array}{l} x \mapsto (\text{left } \underline{0}) \\ y \mapsto (\text{right } \underline{0}) \end{array} \right\} \alpha\text{-typed parts equal}$$

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Succeeds for:

$$\left. \begin{array}{l} x \mapsto (\text{left } _ \alpha) \\ y \mapsto (\text{right } _ \alpha) \end{array} \right\} \alpha\text{-typed parts equal}$$

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

Fails for:

$$\left. \begin{array}{l} x \mapsto (\text{left } \underline{0}) \\ y \mapsto (\text{right } \underline{2}) \end{array} \right\} \alpha\text{-typed parts disequal}$$

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Fails for:

$$\left. \begin{array}{l} x \mapsto (\text{left } _ \alpha) \\ y \mapsto (\text{right } _ \alpha) \end{array} \right\} \alpha\text{-typed parts disequal}$$

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Succeeds for:

$$\left. \begin{array}{l} x \mapsto (\text{right } \neg\beta) \\ y \mapsto (\text{left } \neg\beta) \end{array} \right\} \beta\text{-typed parts equal}$$

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Fails for:

$$\left. \begin{array}{l} x \mapsto (\text{right } \neg\beta) \\ y \mapsto (\text{left } \neg\beta) \end{array} \right\} \beta\text{-typed parts disequal}$$

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Same success/failure results when:

- ▶ Non-variable typed parts are the same
- ▶ Variable-typed parts have the same (dis)equalities

Equality patterns

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \boxed{1} & 0 & 0 & 0 \end{bmatrix}$$

$x \mapsto (\text{right } \underline{2}_\beta)$

$y \mapsto (\text{left } \underline{2}_\beta)$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boxed{1} & 0 & 0 & 0 \end{bmatrix}$$

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$x \mapsto (\text{right } \underline{2}_\beta)$

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x is (right ...)

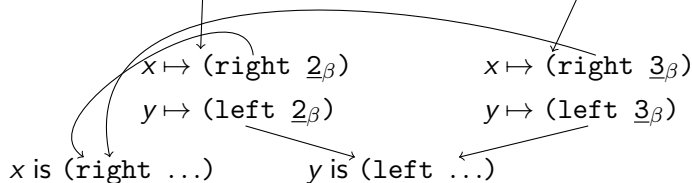
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Equality patterns

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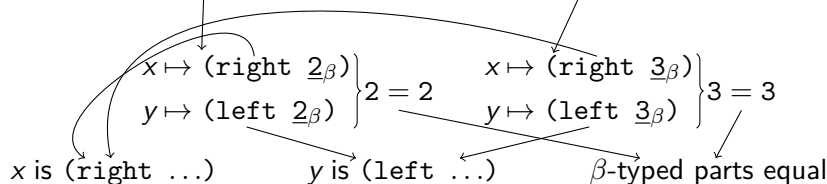
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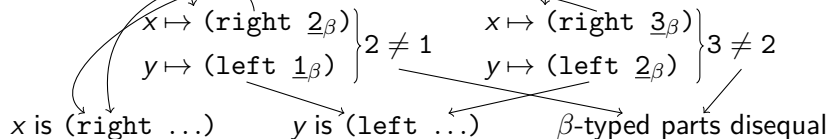
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Equality patterns

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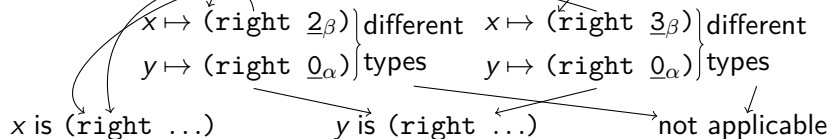
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All done?

Expansion might not work

```
(defrel (has-disequality (y :  $\alpha$ ))  
  (fresh (x :  $\alpha$ )  
    (=/= x y)))
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$$\alpha = \begin{pmatrix} \text{Num } 2 \\ [1 \quad 1] \end{pmatrix}$$
$$\alpha = \begin{pmatrix} \text{Num } 1 \\ [0] \end{pmatrix}$$

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```

$\alpha = (\text{Num } 2)$

$\begin{bmatrix} 1 & 1 \end{bmatrix}$

Always succeeds

$\alpha = (\text{Num } 1)$

$\begin{bmatrix} 0 \end{bmatrix}$

Always fails

Expansion might not work

```
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  (fresh (x :  $\alpha$ )  
    (=/= x y)))
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Always fails

Expansion does not work here.

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```

$\alpha = (\text{Num } 2)$
[1 1]

Always succeeds

$\alpha = (\text{Num } 1)$
[0]

Always fails

Expansion does not work here.

Need to handle small-sized types separately (not “all for one”, but almost).

Compilation

2. Replace polymorphic relation calls with concrete (monomorphic) relation calls
 - ▶ Large-enough calls all use the same relation
 - ▶ Small calls use the correctly-sized monomorphic relation

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All done by generating vanilla semiringKanren code

More details in the paper!

Compiling equal

```
(defrel (equal (x :  $\alpha$ ) (y :  $\alpha$ )) (== x y))
```

(equal 3₄ 3₄) becomes:

```
(fresh ((x1 : (Num 4)) (y1 : (Num 4)) (x2 : (Num
2)) (y2 : (Num 2)))
  (conj
    (== x1 34) (== y1 34)
    (equal $\alpha \mapsto$  (Num 2) x2 y2)
    (disj
      (conj (== x1 y1) (== x2 y2))
      (conj (=/= x1 y1) (=/= x2 y2))))))
```

Introducing semiringKanren

- Goals

- Computing relations

- Types

Polymorphic semiringKanren

- Sum swap

- Expanding relations

- Edge cases and compilation

Conclusion

Represent the (dis)equality structure of relations compactly, and expand as needed.

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Does this generalize?

- ▶ To more than 2 arguments/array dimensions? (hint: yes)
- ▶ To recursive relations? (hint: yes)
- ▶ To more than $==$ and $\neq$?
- ▶ To recursive data types?
- ▶ To non-relational languages?